1. Since the total force induced by the electric and magnetic field is equal to the sum of the individual force induced by each field,
$$\vec{F} = \vec{F}_e + \vec{F}_m = q(\vec{E} + \vec{u} \times \vec{B}),$$
we may solve for $\vec{E}$, the electric field intensity necessary for the particle to maintain a constant speed with no acceleration, by assuming a net zero force on the particle, $F = 0$. Since we know the magnetic field intensity, $\vec{H}$, we may derive the magnetic flux density $\vec{B}$ as
$$\vec{B} = \mu_0 \vec{H} = \mu_0[4\hat{x} + 8\hat{y} + 3\hat{z}]^{\mathrm{T}}.$$
Solving for $\vec{E}$,
$$\begin{aligned}\vec{E} &= -(20\hat{y} \times \mu_0[4\hat{x} + 8\hat{y} + 3\hat{z}]) \\ &= (20)(9.434)\sin(\theta)\ \hat{n},\end{aligned}$$
where θ is the angle between the two vectors and $\hat{n}$ is the normal vector, whose direction is determined with the right-hand rule. We may find the angle using the dot product of the unit vectors $\hat{u}$ and $\hat{y}$, as
$$\theta = \cos^{-1}(\hat{y} \cdot \hat{u}) = \cos^{-1}\left(\frac{8}{9.434}\right) = 0.5586 = 32.01^\circ,$$
and so
$$\vec{E} = -(\vec{u} \times \vec{B}) = -188.68\sin(32.01^\circ) = -\hat{n}\ 100\mu_0.$$
To determine the direction of the normal vector, we may use the determinant definition of the cross product, so that
$$\vec{E} = -\det\begin{bmatrix}\hat{x} & \hat{y} & \hat{z} \\ 0 & 20 & 0 \\ 4 & 8 & 3\end{bmatrix} = -\hat{x}60 + \hat{z}80,$$
and so
$$\vec{E} = (-\hat{x}60 + \hat{y}80)\mu_0 = -\hat{x}[7.54 \times 10^{-5}] + \hat{z}[10^{-4}]\ \frac{\mathrm{V}}{\mathrm{m}}.$$

2. Using the formula for transformer emf,
$$V_{\mathrm{emf}}^{\mathrm{tr}} = -N\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s},$$
we begin by assuming $N = 1$, obviously, and then proceed to solving for $\frac{\partial \vec{B}}{\partial t}$ as
$$\frac{\partial \vec{B}}{\partial t} = \frac{\partial}{\partial t}\hat{x}\ 3\cos(5\pi 10^7 t - \frac{1}{3}\pi y) = -\hat{x}15\pi 10^7 t\sin(5\pi 10^7 t - \frac{1}{3}\pi y)\ \frac{\mu\mathrm{T}}{\mathrm{s}}.$$
Then, aliging the differential surface vector of the loop $d\vec{s}$ with the direction of $\vec{B}$ (the choice of normal vector is arbitrary, but this makes the math a little less complicated) we may express $d\vec{s}$ as
$$d\vec{s} = \hat{x}\ dz\ dy.$$
Plugging these values into our equation for $V_{\mathrm{emf}}^{\mathrm{tr}}$ and using the height and width of the loop as our

bounds of integration,

$$\begin{aligned}
V_{\text{emf}}^{\text{tf}} &= -\int_0^{0.3}\int_0^{0.1} -\hat{x}15\pi 10^7 \sin(5\pi 10^7 t - \frac{1}{3}\pi y)\cdot \hat{x}\ dz\ dy \\
&= \int_0^{0.3}\int_0^{0.1} 15\pi 10^7 \sin(5\pi 10^7 t - \frac{1}{3}\pi y)\ dz\ dy \\
&= \int_0^{0.3} 15\pi 10^6 \sin(5\pi 10^7 t - \frac{1}{3}\pi y)\ dy \\
&= 15\pi 10^6\ [\frac{3}{\pi}\cos(5\pi 10^7 t - \frac{\pi}{3}y)\ \Big|_0^{0.3}\ ] \\
&= 45\times 10^6\ [\cos(1.57\times 10^8 t - 0.3142) - \cos(1.57\times 10^8 t)]\ \mu\text{V} \\
&= 45\ [\cos(1.57\times 10^8 t - 0.3142) - \cos(1.57\times 10^8 t)]\ \text{V}.
\end{aligned}$$

To find current i, we use Ohm's Law and simplify the cosine terms,

$$i = \frac{V_{\text{emf}}^{\text{tf}}}{2R} = \frac{45}{30}\ [\cos(1.57\times 10^8 t - 0.3142) - \cos(1.57\times 10^8 t)]\ \text{A}$$

in the clockwise direction.

3. a) Here, we have another stationary, conducting loop in a time-varying magnetic field induced by a coplanar current-carrying wire. Again using the formula for $V_{\text{emf}}^{\text{tf}}$ and assuming $N = 1$, we may first express the magnetic field induced by a current-carrying wire as

$$\vec{B} = \frac{\mu_0 I}{2\pi R}.$$

Considering that the current in the wire is time-varying and considering the radial distance from the wire in the x direction, we may formulate the magnetic flux density induced by *this* wire as

$$\vec{B} = \hat{y}\ \frac{\mu_0 I_o \cos(\omega t)}{2\pi x}\ \text{T}$$

and its time derivative as

$$\frac{\partial \vec{B}}{\partial t} = \hat{y}\frac{\mu_0 I_0}{2\pi x}\frac{\partial}{\partial t}cos(\omega t) = -\hat{y}\ \frac{\mu_0 I_0 \omega}{2\pi x}\sin(\omega t)\ \frac{\text{T}}{\text{s}}.$$

Then, $d\vec{s}$ may be formulated (in the same direction as $\vec{B}$) as

$$d\vec{s} = \hat{y}\ dx\ dz.$$

Putting everything together,

$$V_{\text{emf}}^{\text{tf}} = \int_S \left[\hat{y}\ \frac{\mu_0 I_0 \omega}{2\pi x}\sin(\omega t)\right]\cdot[\hat{y}\ dx\ dz].$$

Simplifing and pulling out constants,

$$V_{\text{emf}}^{\text{tf}} = \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi}\int_S \frac{1}{x}\ dx\ dz = \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi}\int_0^a\int_{x_0}^{x_0+b}\frac{1}{x}\ dx\ dz.$$

Solving,

$$V_{\text{emf}}^{\text{tf}} = \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi}\left[a\ln\left(\frac{x_0+b}{x_0}\right)\right]\ \text{V},$$

with current flowing in the counter-clockwise direction.

b)